Week 17 Self-Assessment

Research Credibility, p-Hacking, and Replication

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1 Instructions

Test your understanding of research credibility, questionable research practices, and the replication crisis. Understanding these issues is essential for becoming a critical consumer of psychological research.

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2 Part 1: Conceptual Questions

2.1 Question 1: The Replication Crisis

The "replication crisis" refers to:

- (A) Too many studies being published
- (B) Many published findings failing to replicate in subsequent studies
- (C) Researchers copying each other's work
- (D) Difficulty recruiting participants

A famous replication project in psychology found that approximately what percentage of original findings replicated?

- (A) Over 90%
- (B) About 75%
- (C) About 36-40%
- (D) Less than 10%

2.2 Question 2: Why Does Replication Matter?

Replication is important because:

- (A) It shows researchers can copy each other
- (B) It helps us distinguish genuine effects from statistical flukes or methodological artifacts
- (C) It increases the number of publications
- (D) It is required by ethics committees

If a finding does NOT replicate, this suggests:

- (A) The original researchers cheated

- (B) The original finding may have been a false positive, or there may be important moderating factors
- (C) Science is broken
- (D) We should never trust any research

2.3 Question 3: p-Hacking

p-Hacking refers to:

- (A) Breaking into statistical software
- (B) Manipulating data or analyses to achieve $p < .05$
- (C) Using very small p-values
- (D) Reporting exact p-values

Which of the following is an example of p-hacking?

- (A) Pre-registering your analysis plan
- (B) Running multiple analyses and only reporting the significant ones
- (C) Using a large sample size
- (D) Reporting effect sizes

2.4 Question 4: More p-Hacking Examples

Which is a questionable research practice?

- (A) Deciding your sample size before collecting data
- (B) Stopping data collection as soon as you get $p < .05$
- (C) Reporting all your analyses, significant or not
- (D) Using pre-registered hypotheses

Selectively removing outliers until your result becomes significant is:

- (A) Good data cleaning
- (B) Standard practice
- (C) A form of p-hacking
- (D) Always necessary

2.5 Question 5: HARKing

HARKing stands for:

- (A) High Achievement Research Knowledge
- (B) Hypothesising After Results are Known
- (C) Honest And Reliable Knowledge
- (D) Helpful Analyses for Research Knowledge

HARKing is problematic because:

- (A) It takes too long
- (B) It presents exploratory findings as if they were predicted, inflating false positive rates
- (C) It requires too many participants

- (D) Hypotheses are unnecessary

2.6 Question 6: Publication Bias

Publication bias means:

- (A) All studies get published equally
- (B) Studies with significant results are more likely to be published than non-significant ones
- (C) Journals are biased against psychology
- (D) Too many studies are published

Publication bias contributes to the replication crisis because:

- (A) It encourages more research
- (B) The published literature over-represents findings that may be false positives
- (C) It makes journals too selective
- (D) Non-significant results are always wrong

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3 Part 2: Interpretation Questions

3.1 Question 7: Identifying Questionable Practices

A researcher reports: "After extensive data analysis, we discovered an unexpected but significant relationship between X and Y."

What concern does this raise?

- (A) The finding is definitely wrong
- (B) Possible HARKing - the finding may be exploratory but presented as if predicted
- (C) The analysis was too extensive
- (D) The relationship is too strong

3.2 Question 8: Evaluating Research Practices

Which researcher is following best practices?

- (A) Researcher A: Removes outliers until getting $p < .05$
- (B) Researcher B: Only reports significant findings from many analyses
- (C) Researcher C: Pre-registered hypothesis and reports all results regardless of significance
- (D) Researcher D: Changes hypothesis after seeing preliminary data

3.3 Question 9: The "File Drawer Problem"

The "file drawer problem" refers to:

- (A) Researchers losing their data
- (B) Non-significant studies remaining unpublished (in file drawers) while significant ones get published

- (C) Too much data storage needed
- (D) Poor file organisation

This contributes to:

- (A) Better science
- (B) A biased view of what effects actually exist
- (C) Faster publication
- (D) More efficient research

3.4 Question 10: Statistical Power

Low statistical power increases the risk of:

- (A) Finding an effect when none exists
- (B) Missing a real effect (false negative) AND inflated effect sizes when effects are found
- (C) Too many significant results
- (D) Ethical violations

Studies with low power that do find significant effects often:

- (A) Are more reliable
- (B) Report inflated effect sizes (winner's curse)
- (C) Have smaller p-values
- (D) Are more likely to replicate

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4 Part 3: Application Questions

4.1 Question 11: Improving Research Credibility

Pre-registration helps prevent:

- (A) Data collection errors
- (B) HARKing and p-hacking by committing to analyses before seeing data
- (C) Participant dropout
- (D) Plagiarism

Open data and open materials help by:

- (A) Making research faster
- (B) Allowing others to verify findings and detect errors
- (C) Reducing costs
- (D) Guaranteeing replication

4.2 Question 12: Evaluating Claims

A study claims to have found a significant effect ($p = .049$) after testing 20 different hypotheses.

Your main concern should be:

- (A) The effect is very strong
- (B) The sample was too large
- (C) Multiple comparisons increase the chance of a false positive - one in 20 tests would be ‘significant’ by chance
- (D) The p-value is too small

4.3 Question 13: What Makes Research More Credible?

Which features increase your confidence in a finding?

Pre-registered hypothesis and analysis plan: TRUE / FALSE

Large sample size with adequate power: TRUE / FALSE

Only significant results reported from many analyses: TRUE / FALSE

Successful independent replication: TRUE / FALSE

Hypothesis changed after seeing the data: TRUE / FALSE

Effect size reported alongside p-value: TRUE / FALSE

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5 Part 4: MCQ Exam Practice

5.1 Scenario 1: Evaluating a Study

A researcher tests whether a new teaching method improves exam scores. They collect data from 20 students but find p = .08. They then collect 10 more students, re-analyse, and now find p = .04. They report only the final analysis.

1. **What questionable practice is this?**

- (A) HARKing
- (B) Optional stopping / peeking at data
- (C) Pre-registration
- (D) Replication

2. **Why is this problematic?**

- (A) 30 students is too few
- (B) Adding participants until reaching significance inflates false positive rate
- (C) The effect is too small
- (D) They should have used more students initially

3. **A better approach would be:**

- (A) Never collect more data
- (B) Pre-register the sample size and stick to it
- (C) Stop at 20 students
- (D) Use a one-tailed test

5.2 Scenario 2: Interpreting Replication Failure

Study A (original, n=30) finds: d = 0.85, p = .02 Study B (replication, n=200) finds: d = 0.12, p = .31

1. **The replication found:**

- (A) A stronger effect
- (B) The same effect
- (C) A much smaller, non-significant effect
- (D) A negative effect

2. **Which study's effect size is likely more accurate?**

- (A) Study A (original)
- (B) Study B (larger sample, more precise)
- (C) They are equally accurate
- (D) Neither

3. **The original study's large effect size may have been due to:**

- (A) Better methodology
- (B) Winner's curse (inflated effects in underpowered significant findings)
- (C) More careful data collection
- (D) Superior participants

4. **What should we conclude?**

- (A) The original study was fraudulent
- (B) The effect may be smaller than originally reported, or not reliable
- (C) Replications are always wrong
- (D) Never trust psychology research

5.3 Scenario 3: Identifying Best Practices

A researcher plans to test whether exercise improves mood. Before collecting any data, they: - Register their hypothesis on OSF - Specify they will collect exactly 100 participants - State they will use a paired t-test - Commit to reporting all results regardless of significance

1. **This is an example of:**

- (A) HARKing
- (B) p-hacking
- (C) Pre-registration
- (D) Publication bias

2. **This practice helps prevent:**

- (A) Participant dropout
- (B) Changing the analysis plan based on the data
- (C) Recruitment difficulties

- (D) Ethical violations

3. **If the final p-value is .07, the researcher should:**

- (A) Collect more data until $p < .05$
- (B) Change to a one-tailed test
- (C) Report the non-significant result as planned
- (D) Not publish the study

4. **This researcher is demonstrating:**

- (A) Poor planning
- (B) Commitment to transparency and reducing bias
- (C) Unnecessary rigidity
- (D) Overcaution

5.4 Scenario 4: Red Flags in Research

A published paper reports finding significant effects for 18 out of 20 tested relationships (all $p < .05$).

1. **Is this pattern suspicious?** TRUE / FALSE

2. **Why?**

- (A) 18 significant results is impossible
- (B) This pattern is unlikely without selective reporting or p-hacking
- (C) Too many relationships were tested
- (D) The effects must be very large

3. **What would increase your confidence in these findings?**

- (A) If they reported even more significant results
- (B) If the study was pre-registered and the analysis plan showed these specific tests were planned
- (C) If the sample size was smaller
- (D) If only significant results were discussed

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6 Summary Checklist

Before moving on, make sure you can:

- ☐ Explain what the replication crisis is and why it matters
- ☐ Define p-hacking and give examples
- ☐ Define HARKing and explain why it's problematic
- ☐ Explain publication bias and the file drawer problem
- ☐ Describe how pre-registration improves research credibility
- ☐ Identify red flags in research reporting
- ☐ Explain the relationship between statistical power and replication

- ☐ Describe what makes research findings more credible

> **! Becoming a Critical Consumer**
>
> Understanding these issues helps you evaluate research critically. When reading a study, ask: Was it pre-registered? Are all analyses reported? Is the sample size adequate? Has it been replicated? These questions help distinguish robust findings from potential false positives.